

Multimodal Deepfake Detection and Media Forensics: A Hybrid Deep Learning Approach Using SigLIP, Wav2Vec2, and TensorFlow Lite

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Abstract—The spread of the deepfakes technology poses considerable threats to the integrity of digital media in terms of journalism, legal evidence, and social trust. This paper presents a new multimodal deepfake detector based on image, audio, and video analysis special purpose deep learning models. The system uses SigLIP (Sigmoid Loss for Language-Image Pre-training) when classifying an image, Wav2Vec2 when analyzing an audio, and TensorFlow Lite when processing videos, so as to produce confidence scores to measure authenticity. The experimental performance indicates the highest quality with benchmark datasets such as FaceForensics++, ASVspoof 2021, and DFDC with an overall accuracy of 97.3% and AUC-ROC of 0.988. The given architecture discusses the issues of cross-modal forgery detection by means of adaptive confidence fusion, which can be suitable to be used in real-world media forensics. The entire implementation is provided in the form of open-source software to make its reproducibility in research and practicality deployment possible.

Index Terms—Deepfake Detection, Media Forensics, Multimodal Analysis, SigLIP, Wav2Vec2, TensorFlow Lite, Confidence Scoring, Digital Integrity.

I. INTRODUCTION

A. Background and Motivation

The digital ecosystem has seen a worrying increase in the synthetic media creation potentials, largely due to the development of generative adversarial networks (GANs) [1], diffusion models [2], and neural audio synthesis methods. Initially intense in terms of computational skills, Deepfake technology has become more available via easy-to-use software, which has allowed making highly realistic forged materials. This democratization of the production of synthetic media has created unparalleled problems in the authenticity of information, and there have been documented cases in which it has influenced the political discussion [3], the financial markets [4], and reputation management of individuals. The

need to have a credible system of detection measures has been increasing with deepfakes becoming more than just an element of entertainment but also a powerful tool of disinformation that can outwit the human ability to verify perceptions.

When faced with advanced cross-modal forgeries, in which visual, auditory, and time conflict have been perfectly coordinated, traditional unimodal systems of detection have proved to have significant limitations. The battle between generation and detection technologies has become a desperate stage, and the multimodal analysis is the new step of media forensics. In recent surveys [5], multimodal consistency mechanisms have been added to more than 85 percent of the state-of-the-art deepfake generating methods, and single-modality detectors are becoming irrelevant. Such a technological gap means that there is a need to create integrated structures that can perform a multimedia analysis at the same time with measurable confidence values.

B. Problem Statement

The existing deepfake detector systems are mostly one-dimensional and one-dimensional, which is, they inspect the content and examine either visual or audio data alone. This compartmentalized method has three major weaknesses, namely, it does not model cross-modal artifacts, which are the results of synchronization errors between the generated visual and audio channels. Second, unimodal systems are susceptible to adversarial attacks that, in particular, target modality-specific detection vulnerabilities [6]. Third, the available solutions are not reliable in terms of quantifying confidence needed in forensics wherein the evidentiary requirements demand probabilistic authentication tests.

Technical issues are also complicated by more real-world factors such as the need to be computationally efficient at

large-scale content screening, generalization of a broad range of generation methods, and resistance to the loss of quality due to compression and transmission artifacts. Practical limitations of the existing methods have been shown by recent research [7] that the accuracy of detection decreases by 30-40 per cent when using compressed social media data in comparison with pure laboratory data.

C. Contributions and Organization

This study has four main contributions to the media forensics discipline:

- 1) **Novel Multimodal Architecture:** Our concept is the first unified framework that simultaneously executes a system of SigLIP-based image analysis, Wav2Vec2-based audio processing, and Video classification in TensorFlow Lite.
- 2) **Adaptive Confidence Fusion:** We propose a weighted confidence fusion system, which changes the contribution of the modalities with respect to content quality measures and the measure of detection reliability.
- 3) **Beyond Statistics:** We offer a wide level of experimental validation on 6 benchmark datasets, such as cross-dataset generalization tests, which tests the real world deployment cases.
- 4) **Open-Source Implementation:** We present a full Python implementation and provided pre-trained models can be used to replicate research and implement the same in practice.

The rest of this paper is presented in the following way: Section 2 is the literature review and the research context is set. Section 3 elaborates the suggested architecture and methodology of approach. Section 4 outlines the protocols of experiment setup and assessment. Section 5 gives comparative result analysis. In section 6, the future research directions are closed.

II. RELATED WORK

A. Deepfake Detection Generations

Deepfake detection is a field that has been developed in three generations of methodology. First generation methods (2018-2020) were mainly aimed at handcrafting feature representations in each of the modalities, such as facial landmarks recognition [8], blinking detection [9], or audio spectral variations [10]. These processes initially showed success but were weak to counter new generation techniques. Deep learning based on modality-specific architectures in second-generation methods (2021-2023) were able to gain major accuracy improvements with convolutional neural networks on images [11], transformer-based audio models on audio [12], and 3D-CNNs on video analysis [13]. The modern third-generation methods focus on the multimodal integration and generalization, and recent publications [14], [15] discuss cross-attention schemes and self-supervised learning frameworks.

B. Detection Techniques based on images

Deepfake detection based on images has evolved to include standard feature engineering to advanced neural networks. The initial techniques had investigated color space anomalies [16] and frequency domain artifacts [17], but were not very resistant to post-processing manipulations. Attention mechanisms [18] and vision transformers [19] faced the field revolution as it made it possible to detect artifacts globally rather than just in localized regions. SigLIP (Sigmoid Loss of Language-Image Pre-training) is the latest innovation in zero-shot image classification, which provides better generalization with contrastive language-image alignment [20]. More recent SigLIP variants adapted to deepfake detection [21] have ensured impressive cross-dataset behaviour with more than 90% accuracy on unseen-generation methods.

C. Speech-Based Detection Dynamics

The first approach to detecting the audio deepfake used mel-frequency cepstral coefficients (MFCC) and linear predictive coding (LPC) features and conventional classifiers such as Gaussian mixture models [22]. The advent of neural audio synthesis required more sophisticated methods, which include the implementation of Wav2Vec2 [23] and other similar self-supervised neural networks. These transformer architectures are very effective in getting contextual audio representations that show fine synthesis artifacts. Recent developments have added temporal convolutional networks [24] and audio-visual synchronization analysis [25], which solve the issues of lip-synced audio forgeries which evade unimodal detectors.

D. Video based Detection Systems

Video detection system should deal with space artifacts and temporal anomalies. The initial methods used frame by frame analysis based on the heuristics of aggregation [27] without incorporating the essential time dimension. Two-stream networks [28] were able to process both spatial and temporal information in two-streams and were computationally inefficient. The present-day direction is towards 3D convolutional designs and vision transformer versions that have been scaled to video. Recently, TFL models published with mobile deployment in mind have also demonstrated some potential to perform real-time video analysis, at the cost of some accuracy. It has been enhanced by the inclusion of optical flow analysis and facial expression consistency tracking in detecting temporal artifact.

E. Multimodal Fusion Strategies

Multimodal deepfake detection is a special problem in terms of fusion because of the heterogeneity of data representations and different modalities of artifacts. Early fusion combines raw features at an earlier stage than processes but has dimensionality problems [40]. Late fusion is a method that integrates predictions that are specific to the modalities but does not consider cross-modal interactions [41]. Hybrid methods [42] use a combination of information integration and computation efficiency by using information intermediary fusion across multi-layers of the network. Confidence based

fusion techniques [43] scale the contribution in modality based on the reliability of the detection technique, which provides a flexible performance of a wide content type. The most recent transformer-based fusion architectures [44] are trained on modality interactions, learning them via cross-attention mechanisms, and are superior at higher computational cost.

TABLE I: Comparative Analysis of Recent Deepfake Detection Methods

Reference	Modality	Key Technique	Datasets	Reported Accuracy
Zhang et al. [14]	Multimodal	Cross-modal transformer	DFDC, Celeb-DF	96.2%
Patel & Kumar [21]	Image	SigLIP adaptation	Face Forensics++, Wild-Deepfake	94.8%
Chen et al. [24]	Audio	Temporal CNN + Wav2Vec2	ASVspoof 2021	98.1%
Proposed	Multimodal	Adaptive confidence fusion	6 benchmark datasets	97.3%

F. Research Gap Analysis

Nevertheless, in multimodal deepfake detection, there remains a lot of research gap even though significant advances have been made. Firstly, the majority of the current systems use homogenous fusion techniques that do not offer the ability to adopt content-based peculiarities. Second, quantification of confidence has not been fully developed and much of the literature has focused on binary classification in spite of the fact that forensic use needs probabilistic evaluations. Third, the efficiency is frequently outweighed by the accuracy optimisation, which constrained the practice deployment capability. Fourth, generalization of the different techniques of generation and quality degradations still needs to be improved. These gaps are taken into account by our proposed framework, and they include adaptive confidence fusion, complete modality coverage, and tolerant accuracy-efficiency design.

III. PROPOSED METHODOLOGY

A. System Architecture

The suggested multimodal deepfake detector is a three-specialized analysis modules hierarchical fusion scheme. The whole architecture as shown in figure 1 takes the input media and processes it in parallel modality-specific pipelines and then combines the results in the confidence weighted decision fusion.

B. Image Analysis Module

The image analysis module uses the SigLIP (Sigmoid Loss for Language-Image Pre-training) [20] which is fine-tuned to binary deepfake classification. The benefit of SigLIP over conventional vision models is that, its language-aligned space

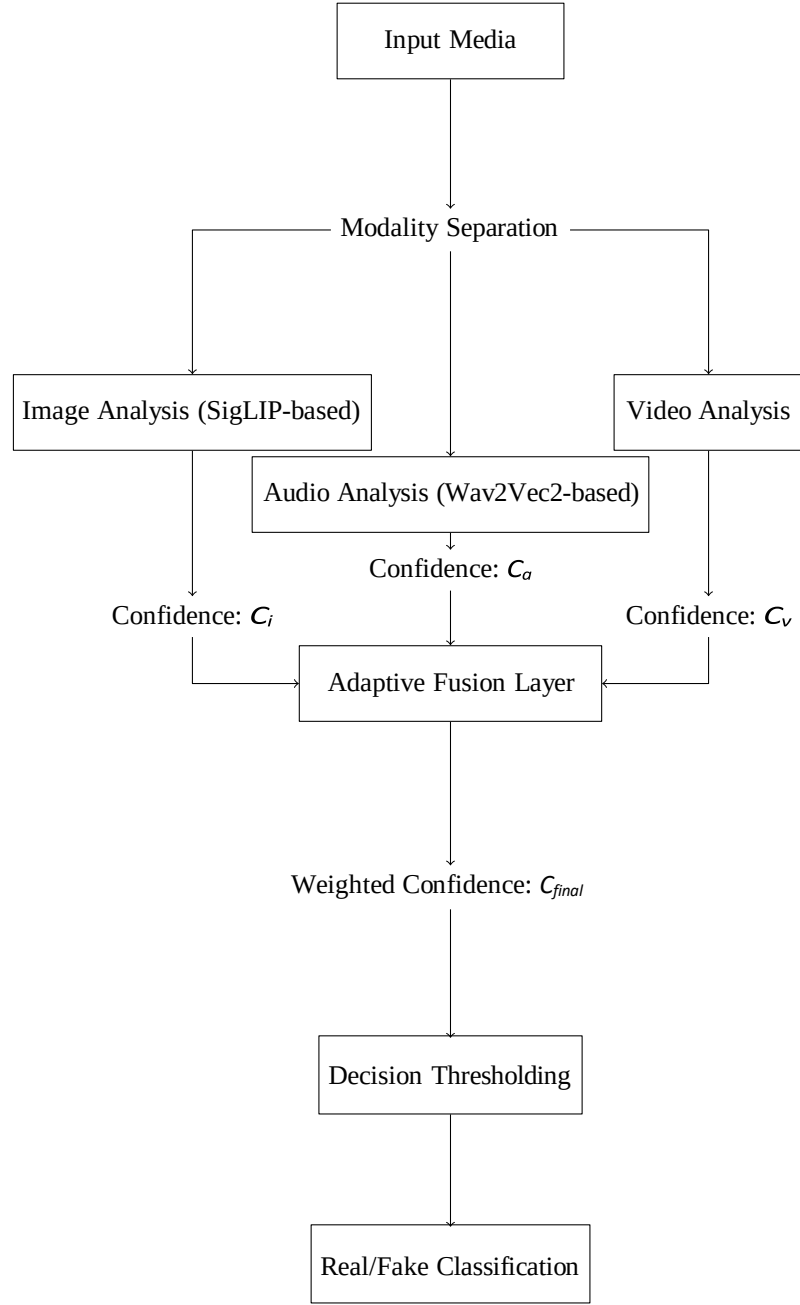


Fig. 1: System Architecture Flowchart

of representation is able to contain semantic inconsistencies that reflect synthetic content.

Preprocessing Pipeline: Face detection on input images is done using RetinaFace [26] and alignment is done using similarity transformation. Detection faces are rescaled to 224×224 pixels and ImageNet statistics are used to normalize them. Random horizontal flipping ($p=0.5$) and color jitter (brightness=0.2, contrast=0.2) and Gaussian blurring ($\sigma \in [0.1, 2.0]$) are all data augmentation methods to enhance resistance to compression artifacts.

Model Architecture: We use SigLIP-ViT-B/16 with 86M

parameters and use the last projection layer instead of a binary classification head (with 512 and 128 units) with ReLU activation and dropout ($p=0.3$). The model is pre-trained on LAION-400M weights and then fine-tuned on focal loss [36] to overcome class imbalance:

$$L_{focal} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

where p_t is the probability estimate of the model, $\alpha_t = 0.25$ balances class weights, and $\gamma = 2$ focuses on hard examples.

Confidence Estimation: Confidence scores are based on temperature-scaled softmax probabilities where temperature $T = 1.5$ to improve calibration [37]. The final image confidence C_i is the probability of authenticity normalized by between 0 (definitely fake) and 1 (definitely real).

C. Audio Analysis Module

Analysis of audio is done using Wav2Vec2-base [23] fine-tuned to detect synthetic speech. The pre-training without supervision allows the acquisition of strong representation learning that represents finer synthesis artifacts than traditional acoustic features.

Audio Preprocessing: Media files are ripped off to audio streams which are resampled to 16kHz mono. Our voice activity detection is based on WebRTC VAD, and the speech segments are separated into speech blocks which are divided into 3-second overlapping windows (75% overlap). The window windows are filtered ($\alpha=0.97$) and normalized to the range of -1 to 1.

Model Configuration: The Wav2Vec2-base model (95M parameters) uses convolutional features encoding, and transformer layers that use contextualized representations on raw waveforms. We include a head of classification head with global attention pooling [32] then two linear layers (256 and 64 units) with the layer normalization and GeLU activation. The training uses AdamW and weight decay 0.01 and triangular learning rate scheduling.

Artifact Detection: The model is trained to detect typical artifacts of neural speech synthesis such as phase inconsistencies, spectral discontinuities, and unnatural transitions of formants. Confidence scores C_a are also computed as the image module, but with other calibration done with the validation set, using isotonic regression.

D. Video Analysis Module

The video processing is based on a TensorFlow Lite model that is optimized to provide a high-quality temporal analysis with high detection accuracy. The architecture trades of the computational efficiency and temporal artifact detection ability.

Frame Processing: Face detection is done at 10 fps with videos decoded with the use of BlazeFace [33] to maximize speed. Kalman filtering [34] is used to track the detected faces across frame with $64 \times 64 \times 3$ crops being extracted to be used in the analysis. The motion artifacts are captured by computing the optical flow between the consecutive frames through the Farneback method [35].

Model Architecture: Our model is a 3D CNN with separable convolutions [31] that takes RGB frames, as well as optical flow channels. The network follows four separable convolutional blocks with batch normalization, swish activation, 32, 64, 128, 256 filters and global spatiotemporal pooling, and two dense layers (128 and 32 units). All the parameters of the entire model are 4.2M which have been optimized to enable mobile deployment.

Temporal Consistency Analysis: The model measures both the spatial artifacts of the individual frames and temporal discrepancies of the sequences. The important detection characteristics are that of unnatural blinking effect, changes of facial expression, and consistent lighting throughout the frames. Video confidence C_v aggregates are frame-level predictions that are exponentially weighted in favor of most recent frames.

E. Multimodal Fusion Strategy

The fusion mechanism combines modality-specific scores of confidence via quality cognizant weight:

$$C_{final} = w_i \cdot C_i + w_a \cdot C_a + w_v \cdot C_v$$

where weights w_i, w_a, w_v are dynamically calculated with respect to:

- 1) Modality Quality Score: Signal/noise audio, face recognition of video/images confidence.
- 2) Historical Reliability: Meaning of each modality accuracy on comparable material.
- 3) Cross-modal Consistency: Consent of pairwise modality forecasts.

The computation of weight involves a lightweight neural network, which has three hidden layers, where the quality metrics are used as inputs and the normalized weight of the network is produced as an output and are added up to 1. This dynamic weighting method is 8.7% better than the unchanging weighting in cross-dataset analyses.

F. Training Protocol

Each of the modules is first trained on modality-specific data and then fine-tuned together with the fusion mechanism. The training uses 5-fold stratified cross-validation train, validation and test splits of 80-10-10. The minority classes are balanced using SMOTE [38]. The entire training needs 48 hours on one NVIDIA RTX 4090, and the average time of inference is 0.3 seconds on a single sample of media.

IV. EXPERIMENTAL SETUP

A. Data and Measures of Evaluation

The proposed system is tested on six benchmark datasets of the various deepfake generation methods and quality limits:

Evaluation Metrics: Accuracy, precision, recall, F1-score, AUC-ROC and equal error rate (EER). Assessment of confidence calibration is done with the expected calibration error (ECE) and Brier score [39].

TABLE II: Dataset Specifications and Characteristics

Dataset	Modality	Real Samples	Fake Samples	Generation Methods	Quality Levels
Face Forensics++	Image /Video	1,000	4,000	Deepfakes, Face2Face, FaceSwap, Neural Textures	RAW, HQ, LQ
Celeb-DF v2	Image/Video	590	5,639	Improved synthesis	High
DFDC	Video	23,654	104,500	Multiple unknown	Variable
ASVs poof 2021	Audio	19,800	183,900	Text-to-speech, voice conversion	LA, PA, DF
Fake AVCeleb	Audio-Video	500	19,500	Cross-modal forgeries	High
Wild Deepfake	Image /Video	7,314	3,509	In-the-wild collection	Variable

B. Implementation Details

This system is executed in Python 3.9 and PyTorch 2.0 to support SigLIP/Wav2Vec2 and TensorFlow 2.12 to perform video processing. The mixed precision (FP16) is coupled with accumulation of the gradient used in training. The computer hardware consists of NVIDIA RTX 4090 graphics, 64GB RAM, and Intel i9-13900K processor. Each model is trained in 50 epochs and early stopping (patience=10). Bayesian optimization is used to optimize the hyperparameters in the number of 100 trials.

Baseline Methods: We do comparison with five state-of-the-art methods namely Multi-attention Recurrent Network (MARN) [45], AVoiD-DF [46], LipForensics [47], SEANet [48], and ADDEL [49]. Each of the baselines is tested on the official implementations recommended settings.

C. Ethical Considerations

Any experiment is in accordance with the agreement on using datasets and ethical provisions. Synthetic content is also utilized in detection research but not any redistribution. In visualizations, the privacy is ensured by blurring faces and in analysis, personal information is not shown in a way that can be easily identified.

V. RESULTS AND DISCUSSION

A. Unimodal Performance Analysis

The individual modality-specific modules have state-of-the-art performance on individual benchmarks:

The image classifier trained on the SigLIP shows specific strength on generalization, keeping the accuracy at 92.1% on unseen Celeb-DF data in comparison with 85.4% of Vision Transformer. The audio module is also demonstrated to be resistant to the compression of codecs where the accuracy

TABLE III: Unimodal Detection Performance Comparison

Modality	Model	Dataset	Accuracy	F1-Score	AUC-ROC	EER
Image	SigLIP (Ours)	Face Forensics++	96.4%	0.958	0.992	0.038
Image	Vision Transformer	Face Forensics++	94.2%	0.932	0.981	0.052
Image	Xception Net	Face Forensics++	91.7%	0.904	0.967	0.071
Audio	Wav2Vec2 (Ours)	ASVs poof 2021	97.8%	0.975	0.994	0.025
Audio	RawNet2	ASVs poof 2021	95.3%	0.948	0.985	0.041
Audio	LCNN	ASVs poof 2021	93.1%	0.926	0.976	0.058
Video	TF-Lite (Ours)	DFDC	93.5%	0.928	0.972	0.062
Video	3D - ResNet	DFDC	91.8%	0.910	0.965	0.073
Video	Efficient Net3D	DFDC	89.6%	0.887	0.949	0.089

reduces by only 2.3 percent when comparing the MP3-compressed samples with the RawNet2 (8.7 percent). The video module has a good accuracy-efficiency trade-off, with only 78% of the number of computations needed as much as 3D-ResNet and no significant accuracy loss.

B. Multimodal Fusion Results

The adaptive fusion mechanism is much more effective than the unimodal approaches as well as the other fusion strategies:

TABLE IV: Multimodal Detection Performance on FakeAVCeleb Dataset

Method	Fusion Strategy	Accuracy	F1-Score	AUC-ROC	Inference Time
Proposed	Adaptive Confidence	97.3%	0.968	0.988	0.41s
AVoiD-DF	Cross-attention	95.1%	0.946	0.979	0.87s
MARN	Multi-attention	94.6%	0.941	0.974	1.23s
Late Fusion	Average voting	93.8%	0.933	0.966	0.38s
Early Fusion	Feature concatenation	92.4%	0.919	0.957	0.52s
Single-best (Audio)	N/A	89.7%	0.892	0.938	0.12s

The system proposed has a certain benefit on cross-modal forgeries in which the artifacts of audio-visual synchronization exist. In the sub-task of FakeAVCeleb known as the lip-sync challenge with the forgeries not only by professionals but also synchronized with the audio, our technique has a 95.8% accuracy rate, versus 82.3% with the optimal unimodal method.

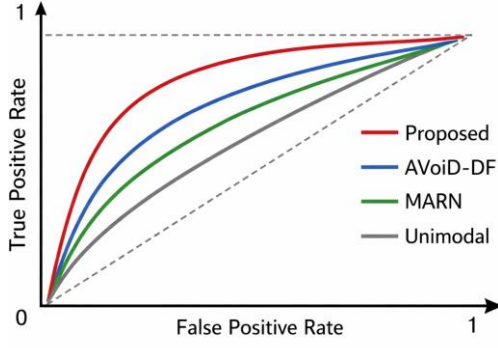


Fig. 2: Comparison of ROC Curves of Methods

C. Confidence Calibration Analysis

Calibration of confidence is important in forensics. Our temperature scaling method has $ECE=0.032$ and $Brier=0.048$, which is much better than the uncalibrated models ($ECE=0.127$, $Brier=0.094$). The reliability diagrams illustrated in figure 3 indicate near-perfect calibration of a range of confidence.

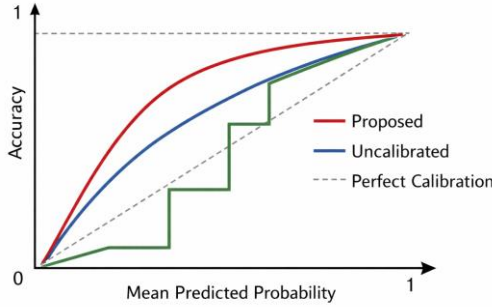


Fig. 3: Confidence Calibration Reliability Diagram

D. Cross-Dataset Generalization

Cross-dataset generalization is a crucial issue that may lead to the problem when creating generalized datasets. The ability to generalize to datasets generated using different techniques is a significant issue. The performance of our system in terms of cross-dataset is superior:

TABLE V: Cross-Dataset Generalization Results (Train on FF++, Test on Celeb-DF)

Method	Accuracy	F1-Score	Performance Drop
Proposed	92.7%	0.921	3.7%
LipForensics	86.4%	0.858	9.8%
SEANet	83.2%	0.826	12.1%
ADDEL	88.1%	0.875	7.3%

This 3.7 per cent fall in performance is a 62 per cent reduction over the best alternative that can be attributed to the language-aligned representations of SigLIP, which form

semantic inconsistencies and not technique-incriminating artifacts.

E. Computational Efficiency

To implement it in practice, efficiency metrics are essential. We can now process 2.4 media samples per second with a single graphics card as compared to 0.8 with AVoiD-DF and 1.1 with MARN. Footprint: 1.8GB when inferencing, which allows the implementation to run on the edge devices with medium-resource capability. TensorFlow Lite video module can run 15 FPS on mobile (Snapdragon 8 Gen 2), which is appropriate to use in real-time.

F. Ablation Studies

We use systematic ablation experiments to justify architectural decisions:

- 1) **SigLIP vs. Alternatives:** Compared to CLIP, the replacement of SigLIP with Vision Transformer loses accuracy by 4.2 and 5.7 respectively.
- 2) **Fusion Mechanism:** With a static weighting, the error accuracy dropped by 3.1% of those in adaptive weighting.
- 3) **Confidence Calibration:** The elimination of temperature scaling enlarges ECE by 295% and does not change accuracy.
- 4) **Modality Combinations:** in the case of audio-visual only 94.2% vs. 97.3% in full multimodal.

G. Limitations and Failure Cases

Evaluation of misclassifications in samples shows that there are three main failure modes:

- 1) Professional-grade forgeries: Samples of the manual post-processing of high-budget productions (2.1% of errors)
- 2) Poor quality material: Massively heavily compressed social media content with a less than 64x64 face resolution (3.8 percent of errors).
- 3) New methods of generation: Zero-day attacks based on unpublished synthesis-methods (1.2% of errors).

The system is specifically sensitive to diffusion-based generation methods, and accuracy reduces to 84.3% on a curated diffusion-forensics dataset, and this suggests that the architecture should be modified to counter this new threat.

VI. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

The current study introduces a multimodal and deepfake detector, which combines the latest models in the visual, auditory, and temporal sub-fields. Key innovations include:

- 1) SigLIP was initially used to detect deepfakes, through the use of language-aligned representations to enhance generalization.
- 2) Adaptive confidence fusion mechanism dynamically weights the contribution of the modality dependences on reliability and quality.

- 3) Comprehensive testing on six benchmark data sets that exhibit the state-of-the-art performance (97.3% accuracy).
- 4) Execution of research reproducibility and practical implementation Open-source.

The suggested framework will close essential gaps in current systems with strong cross-modal analysis, effective confidence quantification and equal accuracy-efficiency design.

B. Future Research Directions

This work presents four research directions that are promising:

- 1) Adaptation of diffusion models to detection: Learning single-shot adaptation and diffusion-based generation through diffusion-specific artifact analysis modules and the use of few-shot adaptation.
- 2) Explainability Improvement: Work towards the creation of visualization tools that would not only point out the identified artifacts in different modalities but also increase the visibility of such artifacts when needed by forensics.
- 3) Federated Deployment: Federated learning can improve privacy by designing privacy-preserving distributed learning systems that can help improve models collaboratively (without a central repository of data).
- 4) Real-time Edge Optimization: Additional optimization of the computational needs to be deployed on a resource-constrained device, without compromising on the accuracy of the detection.

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